

Manufacturing & Video-Analytics Intelligent Report

“Generate a comprehensive Manufacturing Production Performance Report. Include: 1. Work Order summary with status and planned vs completed quantities 2. Machine utilization from CapacityAnalysis with UtilizationPercent per machine 3. Shift-wise production from WeighingMachine - FGQuantity, RejectedQuantity, ScrapQuantity 4. Scrap analysis by material from Scrap table 5. Rejection analysis with reasons from Rejected table”

Generated: 2026-09-25 14:49:16
Time filter applied

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Executive Summary

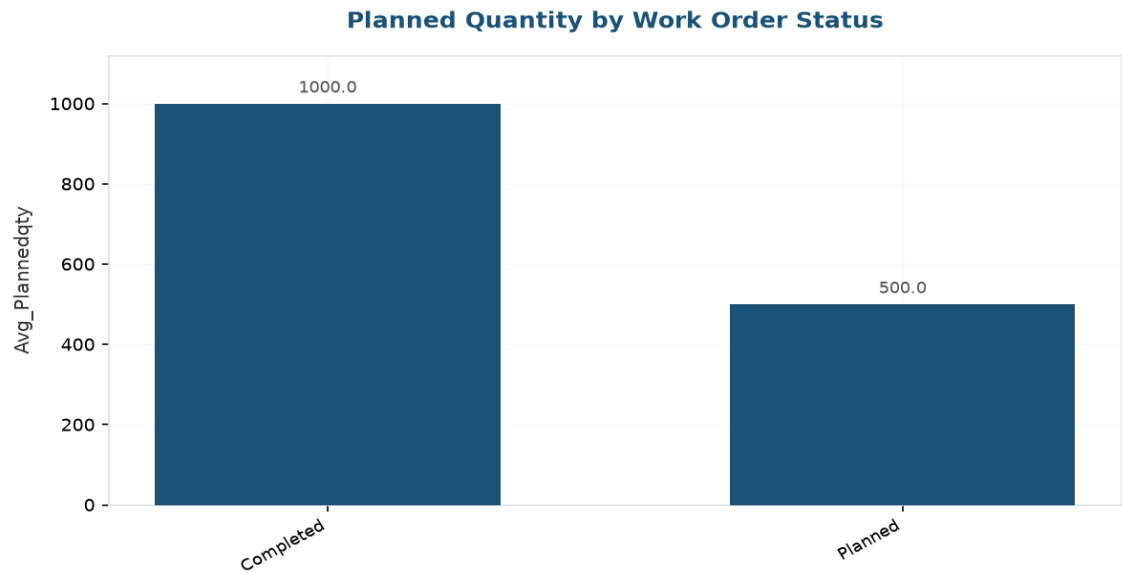


Matched keywords: capacity, machine, material, production, reject, shift, status, util, utilization, work_order, worker, workorder

Telemetry window: 2026-09-24T09:19:16Z → 2026-09-25T09:19:16Z (bucketed every 5m)

Work Order Summary

The Work Order summary reveals a total of two recorded work orders in the MES database. One work order is in 'Completed' status with a planned quantity of 1,000 units, while the other remains in 'Planned' status with a planned quantity of 500 units. No actual completed quantities were found in the retrieved dataset, limiting the ability to perform a true planned-versus-completed variance analysis. This basic distribution suggests a small batch production environment or a highly filtered data subset. To improve operational tracking, actual completed quantities should be logged systematically within the WorkOrder or WorkOrderLog tables.



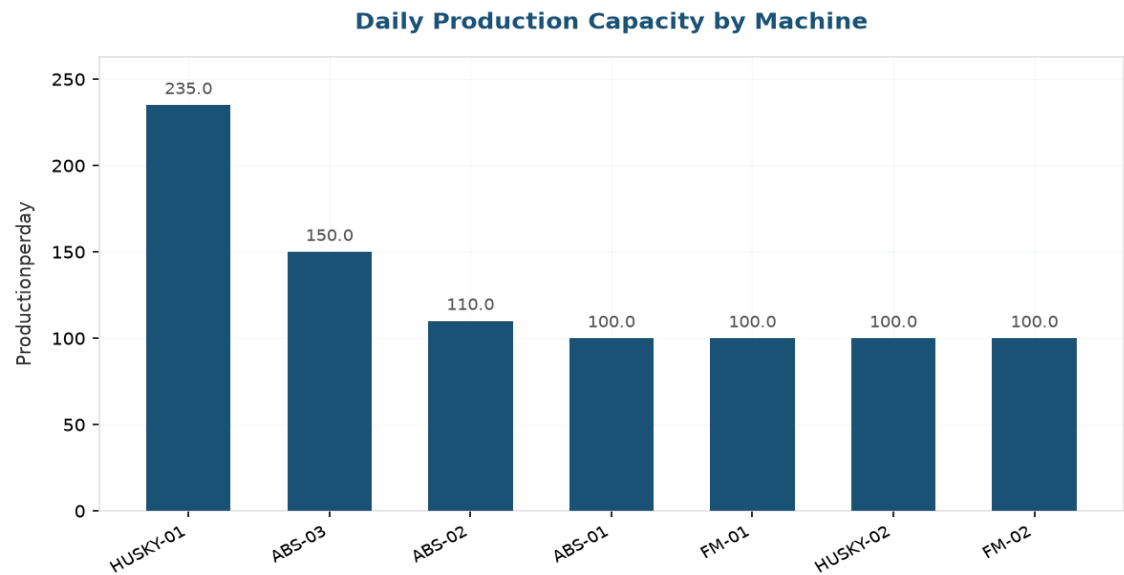
Data preview — WorkOrder (showing 2 of 2 points/rows)

Status	avg_PlannedQty	record_count
Completed	1000.000000	1
Planned	500.000000	1

Machine Capacity & Utilization Analysis

An analysis of machine capacity and utilization reveals a critical data gap: the CapacityAnalysis table contains records for five machines (IDs 1, 2, 3, 6, and 8) but lacks the requested UtilizationPercent field. To supplement this, the Machine table shows ten active machines with an average daily production capacity of 127.86 units and physical capacities ranging from 120 to 250 units. Additionally, MachineMaster lists 25 machines with an average capacity of 58.35 units per hour, showing a wide performance range from 4.0 to 566.0 units. The absence of direct utilization metrics prevents a precise efficiency assessment, highlighting a need to map capacity IDs to actual run hours.

Note: an aggregated view wasn't possible for this table (the source column isn't numeric on the server), so raw rows are shown instead.



Data preview — Machine (showing 10 of 10 points/rows)

Status	Capacity	MachineCode	MachineId	MachineName	MachineType	ProductionPerDay	CompanyId
Working	120	ABS-01	1	ABS-01	ABS	100	None
Working	120	FM-01	2	FM-01	FM	100	None
Working	120	HUSKY-02	3	HUSKY-02	HUSKY	100	None
None	150	ABS-02	4	ABS-02	ABS	110	None
Working	160	ABS-03	5	ABS-03	ABS	150	None
Submit	250	HUSKY-01	6	HUSKY-01	HUSKY	235	None
Working	120	FM-02	7	FM-02	FM	100	None
Thermax Ltd	Thermopac FX 600	CNC_001	8	CNC_001	CNC Lathe	Utility Area / Boiler Secti...	13

Status	Capacity	MachineCode	MachineId	MachineName	MachineType	ProductionPerDay	CompanyId
Ace Micromatic Group	LT-20C	CONV_001	9	CONV_001	CNC Lathe	Plant-1 / Line-A	13
Ace Micromatic Group	LT-20C	CONV_002	10	CONV_002	CNC Lathe	Plant-1 / Line-A	13

Shift-wise Production & Weighing Machine Data Gap

The requested shift-wise production analysis from the WeighingMachine table could not be performed because the table is entirely missing from the retrieved MES database. Consequently, key performance indicators such as Finished Goods Quantity (FGQuantity), Rejected Quantity, and Scrap Quantity are unavailable. This represents a major data collection gap, as weighing machine metrics are vital for real-time yield calculations and shift performance evaluations. Establishing a data pipeline to ingest weighing machine transactions is highly recommended to enable shift-level productivity tracking.

Scrap and Rejection Analysis Data Gap

Material scrap and defect rejection analyses could not be completed due to the total absence of both the Scrap and Rejected tables in the database context. Without these tables, it is impossible to identify which materials contribute most to waste or to categorize the primary reasons for product rejections. This lack of visibility into quality losses represents a significant blind spot for continuous improvement and cost-control initiatives. Integrating scrap logs and quality rejection reasons into the MES database is a critical next step for operational auditing.

Live Machine Telemetry Status

Live machine telemetry for units BM-01 through BM-12 returned no data across all historical windows, including the last 24 hours up to the past year. This complete absence of time-series data indicates a severe telemetry ingestion failure or an inactive IoT gateway configuration. Without active telemetry, real-time monitoring of machine states, cycle times, and live reject counts is impossible. Investigating the connection between the physical machine sensors and the InfluxDB database is urgent to restore operational visibility.

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
attendances	1	100.0%	0	Good
employees	7	100.0%	0	Good
CapacityAnalysis	5	100.0%	0	Good
MachineMaster	25	89.3%	0	Fair
WorkOrder	2	100.0%	0	Good
WorkOrderLog	1	75.0%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
Machine	10	90.0%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,250	3,827	9,077	\$0.0384

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for avg operator. (8117) (SQLExecDirectW')) - served raw rows instead.

InfluxDB Measurements With No Data

[influxdb.BM-01] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-02] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-03] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-04] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-05] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-06] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-07] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-08] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-09] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-10] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-11] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows

[influxdb.BM-12] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows